Detecting Brain Hemorrhage in CT Images with ResNeXt101

Bachelor's Project Report

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Abstract

This project uses a trained image model to detect whether a head CT slice contains hemorrhage. A slice is one cross-sectional image from a CT scan. The selected model is ResNeXt101 with a small channel-attention module called ECA. Images are prepared in a consistent way and passed through the network. The report focuses on the model's Any output, which indicates the presence of hemorrhage.

The test evaluation includes 74,950 slices. It reports 95.22% accuracy, 75.80% sensitivity and an F1 score of 81.71% for Any. The results show useful detection performance, but the model still misses some positive slices. [1, 2]

1 What the project does

A CT scan contains many slices. Each slice shows a different cross-section of the head. In this project, the model looks at one slice at a time and produces a score for whether hemorrhage is present. It does not use the neighboring slices to make that decision.

The task is image classification: assign an image to a category. Here, the two outcomes of interest are hemorrhage present and hemorrhage absent. The model is not asked to draw the boundary of the bleeding, measure its volume or recommend treatment.

The project uses transfer learning. This means starting with a model that has already learned useful image patterns, rather than starting with random weights. Those learned patterns provide a starting point for adapting the network to CT images. They do not mean that the original pretrained model already understood hemorrhage.

The work is organized around three steps: prepare the images, obtain the network's prediction and compare that prediction with the known label. The main question is how well the selected model detects positive slices. A second question is whether its high overall accuracy gives a complete picture of its behavior.

The scope contains one CNN. It does not include feature reduction, a separate tree classifier, a recurrent network or a comparison of several architectures. ResNeXt101 remains the backbone throughout the project.

2 Data and Image Preparation

The images and their labels

The images come from the RSNA 2019 intracranial hemorrhage dataset. A label is the recorded answer associated with an image. For the Any label, 1 means hemorrhage is present and 0 means it is absent. The saved test evaluation contains the following slices. [1, 2]

Table 1. Test slices in the saved CNN evaluation.

Test group	Slices
Hemorrhage absent	64,392
Hemorrhage present	10,558
Total	74,950

Most test slices are negative. This matters because a program that always predicts no hemorrhage would already be correct on about 85.91% of this test set. We therefore need to measure how many positive slices the model finds, as well as how often it is correct overall.

The data have separate training, validation and test partitions. Training data are used to learn the model's weights. Validation data are used to check choices during development. Test data are used for the final assessment. The supplied splitting code groups images by imaging series. Patient overlap should also be checked, because one patient may have several series.

Preparing a slice for the network

The images are stored as DICOM files. DICOM contains the image and information needed to read its pixel values correctly. The loader uses that information to convert the stored values to Hounsfield units, the intensity scale used for CT images.

Next, the code creates three views of the same slice using different intensity ranges: brain, subdural and soft tissue. This process is called windowing. It is similar to choosing different contrast settings so that different details are easier to see. These are three views of one slice, not three separate scans.

The three views are stacked as the input image's three channels. The image is resized to 256 by 256 pixels. Its values are then normalized using the channel means and standard deviations specified in the code. Normalization puts the input values on a consistent scale. [2]

During training, small random image changes provide additional variation. Test images use fixed preparation without random changes. Keeping the preparation consistent is important: a trained model may behave differently if the channel order, image size or intensity handling changes.

3 How the Model Works

Learning useful image patterns

ResNeXt101 is a convolutional neural network. A convolutional network uses learned filters to find patterns in an image. Earlier layers can describe simpler patterns, such as edges and changes in intensity. Later layers combine this information into more complex features. A feature is a numerical description learned from the image; it does not need to have a separate medical name.

The selected implementation uses the pretrained resnext101_32x8d_wsl backbone. It also includes ECA, a small module that adjusts the importance of feature channels. These internal channels describe patterns learned by the network and are different from the three channels of the input image. The ECA output is added to the original features before the final classification stage. [2, 3]

The network then averages the feature maps to produce a summary of the slice: a vector of 2048 numbers. These numbers are learned image features, not 2048 pixels or disease categories. A final linear layer turns them into six raw scores.

Using the Any output

The existing model has five outputs for hemorrhage subtypes and one output called Any. This report uses Any to answer the simpler question: is hemorrhage present? The other outputs remain in place so that the model is compatible with its existing weights. This is not a newly trained model with only one output.

A raw score is called a logit. A sigmoid function converts it to a value between zero and one. A threshold then turns the score into a yes-or-no decision. The exact threshold matters because changing it can change the number of detected cases and false alarms.

Training and running the implementation

The code uses Python and PyTorch. Its visible training settings include batches of 32 images, the Adam optimizer and a learning rate of 0.00001. A batch is a group of images processed in one training step. The loss function, BCEWithLogitsLoss, measures disagreement between the raw outputs and the labels so that the weights can be updated.

The evaluation stage loads the trained ECA model checkpoint and applies it to the test set. A checkpoint stores the learned model weights, while an epoch is one complete pass through the training data. To reproduce the same evaluation, the corresponding checkpoint, dataset split and preprocessing settings must be used. The numerical results below summarize the test evaluation for the Any output.

4 Results and Conclusion

Reading the results

The test evaluation reports the following values for Any on 74,950 slices. Precision, sensitivity and F1 describe the positive hemorrhage class. [2]

Table 2. CNN test results.

Measure	Value
Accuracy	95.22%
Precision	88.62%
Sensitivity	75.80%
F1 score	81.71%
AUC	0.9689

For illustration, a sensitivity of 75.80% means roughly 76 out of every 100 positive slices are detected, while about 24 are missed. Precision answers a different question: among 100 positive predictions, roughly 89 are correct. These are examples of the ratios, not exact test counts or patient counts. Accuracy counts all correct decisions; F1 combines precision and sensitivity.

The evaluation compares raw logits with 0.5, equivalent to a sigmoid probability threshold of about 0.6225. Changing this decision rule requires a new evaluation. AUC measures score ranking rather than the percentage of correct decisions.

What the result tells us

The selected CNN provides useful separation between positive and negative slices in the saved experiment. However, high accuracy does not mean every hemorrhage is detected. The lower sensitivity is an important limitation. The model also processes slices independently, and its performance on an independent hospital dataset has not been established.

For reproducible evaluation, the same checkpoint, dataset split and preprocessing settings should be used. Patient separation between data splits should also be checked. ResNeXt101 remains computationally demanding.

References

[1] Rahimi, P. Detection and Classification of Brain Hemorrhage in CT Images Using Deep Learning. Shahid Chamran University of Ahvaz. The broader study uses RSNA Intracranial Hemorrhage Detection data, 2019.

[2] Project implementation: Shaghayegh_Shushtari_Bachelor_Project.ipynb.

[3] Xie, S., et al. Aggregated Residual Transformations for Deep Neural Networks. CVPR, 2017. Wang, Q., et al. ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks. CVPR, 2020.